Kumar Abhinav

GenAI Software Engineer
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Core Technologies & Skills

Languages & Frameworks:
Python · Java · TypeScript · FastAPI · Spring Boot · Angular

APIs & Integrations: RESTful APIs · Temporal · Azure AI Search · SAP, Azure OpenAI

AI & Automation: OpenAI API · LangGraph ·Transformers ·RAG· Agentic AI · MCP · LLM Eval

Data & Analytics:
SQL(Advanced) · Elasticsearch · Tableau · AWS Redshift

AI Software Engineer with 4+ years of experience building production-grade GenAI and agentic AI systems using Python, FastAPI, LangGraph, RAG, Pinecone, and Temporal. Delivered scalable AI products that improved query latency by 50%, achieved 65% adoption in 2 months, and saved 2,000+ engineering hours annually while supporting reliable enterprise workflows. Combines software engineering depth with 12+ years of prior analytics and supply chain domain to build AI systems that are technically strong and business impactful.

Career Highlights

- **Production GenAI Systems:** Built and deployed agentic AI and RAG solutions using LangGraph, Pinecone, GPT-4, and FastAPI, reducing triage/query handling time by up to 45% and driving 65% adoption.
- **AI Product Engineering:** Shipped customer-facing and internal AI products across utilities, healthcare, and pharma, delivering 25–40% efficiency gains and supporting ~$1M in upsell / savings impact.
- **LLM Engineering Project:** Built a GPT-style transformer (15.2M parameters) from scratch in PyTorch with optimized training and interactive generation workflow.

Work Experience

AiDash | Sr. Software Engineer (Full Stack), Palo Alto, CA. Dec 2025 – Present

AiDash is an AI-first vertical SaaS company helping power utilities become more resilient and sustainable using satellite intelligence and AI. I work on the Professional Services team, partnering with customers and Customer Success to build custom product features and GenAI-driven solutions.

Specific Contributions:

New Products (0⟶1)

- **New Internal Product | Agentic Code Generation Platform —** Built the first stage of a multi-agent engineering workflow that ingests Jira tickets (including comments and attachments), generates implementation plans, routes them through reviewer agents, and supports human approval before triggering downstream frontend, backend, QA, branch-creation, runbook, and Jira-update workflows. **Improved code-generation** accuracy to **~80–85%, reduced trivial-ticket deployment time** from 7 days to 2 days **(~70% faster**), helped reduce backlog, and **avoided** the need for **two contractor hires**, **saving ~$50K annually**.

Customer-Facing Product Features

- **New Product Feature | NLP Smart Filtering | Reduced operator filter setup from 2–3 minutes to under 10 seconds**, improving wildfire-response workflows by ~60% through natural-language querying over complex utility infrastructure data. **Built an end-to-end React/TypeScript + FastAPI flow using LLM-based intent parsing**, deterministic hierarchical filter execution (state ⟶ region ⟶ circuit), and real-time map/metrics refresh with validations guardrails.
- **New Product Feature | ARIA — Autonomous Risk Intelligence Assistant |** Built an **agentic AI** assistant for alpha testing that enables 24/7 proactive monitoring across 10,000+ circuits, **reducing manual oversight by ~80%** and accelerating mitigation decisions through auditable alerts and recommendations. **Implemented LangGraph + MCP multi-tool orchestration** for circuit telemetry, risk signals, and reporting, with structured outputs, retries/timeouts, activity logging, and human-in-the-loop.

Internal Productivity & Ops Automation

- **Internal Automation | Jira Ticket Intelligence (Rovo + Jira Automation)** | **Decommissioned an external triage tool** by automating Theme classification + Priority Score recalculation for new/updated tickets, improving prioritization consistency and sharply reducing manual grooming effort for CS + Engineering. **Implemented Atlassian native Rovo agents + Jira Automation** with a strict JSON-only contract, normalized scoring over Customer Impact/Required-By/ORBIT/VIBE, and hardened reliability via field validations, change triggers, and audit-log tracing to prevent malformed payloads/invalid option labels.

- **CCP Notification Service:** Built a tenant-aware, event-driven **notification microservice** using **FastAPI, PostgreSQL, Django/DRF, and React**, enabling in-app bell/inbox and HTML email notifications for job lifecycle, assignment, approval, and onboarding events. Designed idempotent event ingestion, secure machine-to-machine APIs, deep-link routing.

LinkedIn | Sr. Software Engineer (Contractor), Sunnyvale, CA. Oct 2024 – Dec 2025

Own and ship full-stack features for LinkedIn's internal Employee Productivity Platform, serving 28K+ employees. Partner with Product, UX, and IT Service Management (ITSM) to define roadmaps, build APIs, and iterate on user feedback—delivering measurable impact.

Specific Contributions:

- **Product Development – Employee Productivity Platform: Impact – Saved ~$200K/year in infra (VM shutdown) and freed ~1 FTE; reduced MTTR by ~45% with 65% adoption in 2 months.** Led end-to-end build of an **internal GenAI chatbot** (Azure OpenAI GPT-4/GPT-4o + RAG on Azure AI Search & Glean) to auto-answer employee and IT queries, migrating legacy Java services to Python/FastAPI and simplifying platform.
- **Product Features – ServiceNow, Contract & Finance Integrations: Impact – Cut low-value ticket creation by ~30% and prevented ~$50K+ in potential revenue leakage.** Integrated ServiceNow with the chatbot for incident deflection, built Contract Finder (Salesforce + Dynamics), and Franklin Finance PO lookup so support and finance teams could self-serve contract and payment details.
- **Data Pipeline & Engineering Excellence: Migrated employee data pipelines: Impact – Eliminated ~95% pipeline failures and saved 416 hrs/year ($65K at $160/hr)** of engineer time spent on debugging and manual fixes**.** Migrated LDAP/Slack user refresh, Cosmos priming, and MS Graph working-hours pipelines into **Temporal workflows** and hardened CI/CD with dependency cleanup and monitoring.
- **AI-Powered Test Generation Platform: Impact –** Cut test-data prep from **days to ~90 seconds per run** with 100% FK integrity **at only $0.20–$0.50/run**. **· Designed an LLM-driven platform** (Python, LangGraph, Oracle) that turns business documents into executable test scenarios and FK-aware synthetic test data for faster QA Validation.
- **Marketing Brief & Asset Automation (BP / M&C / JLR): Impact –** Automated JSON payload creation for resize/translation workflows, **reducing manual marketing ops effort** and enabling downstream agentic execution**. Built a folder-scanning and intent-detection pipeline (LangGraph + Azure OpenAI) to parse briefs, classify assets, read Excel translation sheets,** and generate downstream JSON (translation, text replacement, resize) aligned to existing production schemas.

RFXCEL | Software Engineer, San Ramon, CA. May 2023 – Dec 2024

Partnered with Sales Engineering to architect and ship new product features, analytics and automation products for healthcare & pharma clients like Cardinal, AbbVie and Walgreens to driving efficiency, scalability, and user adoption.

Specific Contributions:

- **Internal Productivity:** Leveraged GenAI to prototype and launch several tools for internal productivity automation such as **Text2SQL Query Assistant**: Built a GenAI-powered tool using **OpenAI's API and AWS Redshift** to translate **natural-language prompts into SQL queries** for Customer Support Analysts. Reduced support-ticket **processing time by 45%,** reduced the dependency to hire three contractor roles, and **saved $100 K annually. Automated Release-Notes Generator:** Developed a pipeline leveraging the JIRA API, Confluence data, and OpenAI to draft customer-friendly release notes automatically at the end of each sprint. **Saved 576 Engineering Working hours annually** and increased customer satisfaction by 65%.
- **Product Development – Real-Time Inventory Lifecycle Dashboard:** Built a SQL⟶ Elasticsearch pipeline feeding an Elasticsearch index and Kibana dashboard to deliver on-demand, country-level inventory and supply-chain insights. Reduced **data-query latency by 50%, boosted** operational efficiency **by 25%,** upsold the existing traceability platform to pharma clients, and **generated $ 1 M in additional revenue**. Empowered government regulators, suppliers, and manufacturers with actionable, real-time data
- **Product Development: Legacy product enhancement: two features: (a) Escalation management:** Developed Java Spring Boot REST APIs and an Angular UI to enable Walgreens and its distributors to configure exception rules in real time. Empowered users to define alerts on the fly—cutting **resolution time by 40%** and **reducing operational losses** (support staffing and logistics) **by approx. $ 1 M / year**. **(b) Automated real-time inventory & order synchronization:** Developed a Java Spring Boot public API to integrate RfXcel's platform with Manufacturer (AbbVie) SAP system—**automating 100% of order handoffs**, eliminating manual, error-prone updates, ensuring 100% accurate, up-to-date inventory and order status, and **saving $ 200 K / year in labor costs**.
- **Jira Conversational Assistant (agentic AI) — 10× faster triage**: Built a LangGraph-based assistant that turns natural-language queries into ticket actions and escalation-risk scoring. **Triage dropped from ~30–40 min to <3–4 min per inciden**t (order-of-magnitude), **preventing ~20 escalations/month** and saved **~16 engineer-hours per week (~832 hours/year),** cutting **support costs by ~$120 K annually.**

Portfolio – https://abhinav0905.github.io

Early Career | Staff - Business Data Analyst, Chennai/Mumbai, India. Apr 2009 – Aug 2022

Owned and delivered end-to-end supply-chain optimization products, combining SAP automation, advanced analytics, and forecasting to drive compliance, cost savings, and operational efficiency—culminating in successful McKinsey audits and $1 million in working capital improvement.

Specific Contributions:

- **Requirements & Stakeholder Management:** Led requirement gathering with Operations, Finance, Procurement, and Plant leadership for inventory and cost-visibility initiatives; converted business pain points **into BRD/FRD**, user stories, and acceptance criteria for reporting and process improvements.
- **Data Mapping & Master Data Alignment (SAP):** Created data dictionaries and mapping sheets linking equipment → spare parts → consumption history across plants; standardized material classification and attributes in SAP to enable consistent reporting and inventory segmentation.
- **Global Inventory Optimization Engine:** Designed and built a quantitative segmentation framework (JIT, VMI) integrated into SAP —leveraged Python and SQL to map equipment to spare parts across plants, enabling accurate inventory classification **reduced global inventory by 25%, freed up $1 M in working capital.**
- **Manufacturing Analytics & Loss-Reduction Module**: Developed a Python-based predictive analytics product leveraging linear regression machine learning model, hypothesis testing, and variance analysis to pinpoint production inefficiencies. Presented Executive **Tableau dashboard** for the leadership group. **Eliminated 22%** of manufacturing losses, and **reduced manufacturing cost from $200/MT to $150/MT.**

Patents

- **Provisional Patent – Embedded Agentic Retrieval with Resettable Continual LoRA Adapters (USPTO 63/967,742, 2026)** Designed a RAG architecture that embeds a small LoRA-adapted model into the retrieval layer to perform query rewriting, reranking, and context packing, reducing LLM calls, token cost, and latency while improving multi-tenant, provenance-aware retrieval.
- **Provisional Patent – Proof Carrying Workflow Provisional Filing (USPTO Application No. 64/083,845, 2026) –** A framework for verifiable AI agent execution: cryptographic "tool receipts," hash-chained tamper-evident logs, deterministic replay, and two-phase gating before irreversible actions.

Publications

Authored and submitted an IEEE research paper on a career-aware resume-tailoring system using 12-node LangGraph pipeline, multi-source RAG, provenance tracking, and anti-hallucination guardrails; pilot evaluation across 9 job descriptions showed an average ATS-fit improvement of +7.8 points for domain-aligned roles.

Personal Projects

Small Language Model from Scratch – Architected and trained a **GPT-style transformer (15.2M parameters)** in PyTorch from scratch. Implemented complete **training pipeline on Apple M2 GPU** with custom dataset preprocessing and batching. **Achieved 97% accuracy (perplexity: 1.03)** on domain-specific text generation task. Built interactive generation interface with temperature and top-k sampling controls. Optimized training with gradient accumulation, mixed precision, and thermal management for consumer hardware.

Demo Video - https://www.linkedin.com/feed/update/urn:li:activity:7382074871822032896/

Education and Academic Background

M.S. in Business Analytics, Aug 2022 to Dec 2023 || **California State University East Bay**, Hayward, CA

B.Tech. in Chemical Engineering (Major) | **Computer Science & Engineering (Elective),** Aug 2005 to Dec 2009 || **Sathyabama Deemed University** Chennai, India

Academic Recognition: Fully Funded Ph.D. Admission Offer – **Computer Science, University at Buffalo (SUNY),** Department of Computer Science and Engineering; selected to work under Dr. Venu Govindaraju. Listed as an academic achievement; current focus remains full-time industry roles.

Portfolio – https://abhinav0905.github.io